

40.520 Stochastic Models: HW-5. Martingales.

1.) $X_{n+1} = \begin{cases} \frac{1}{3} + \frac{2}{3} X_n & \text{with prob } X_n \\ \frac{2}{3} X_n & \text{with prob } 1 - X_n \end{cases}$

RTP: This DT Markov is a martingale.

Solution

Sufficient condition for martingale:

$$E[X_{n+1} | X_n] = X_n$$

$$\begin{aligned} E[X_{n+1} | X_n] &= \left(\frac{1}{3} + \frac{2}{3} X_n\right) X_n + \frac{2}{3} X_n (1 - X_n) \\ &= \frac{1}{3} X_n + \frac{2}{3} X_n^2 + \frac{2}{3} X_n - \frac{2}{3} X_n^2 \end{aligned}$$

$$E[X_{n+1} | X_n] = X_n \quad \rightarrow \text{proof!}$$

2.) Let $\{X_n, n=1, 2, \dots\}$ be martingale.

RTP: For any set of $k < l$, X_k is uncorrelated with X_l . i.e. $E[(X_l - X_k) X_k] = 0$

Solution

Since $k < l$, X_k is already known by time l .
 $\therefore$ when condition on all info upto time l ,
 X_k is a constant term.

$$E[(X_l - X_k) X_k] = E[X_k E[(X_l - X_k) | \text{information upto time } l]]$$

tower rule.

Now, since $m > l$.

$$E[(X_m - X_l) | \text{information upto } l] = E[X_m | \text{info}] - X_l$$

Also since X_k is a martingale.

$$E[X_m | \text{info upto time} = k] = X_k.$$

$$\therefore E[X_m | \text{info}] - X_k = 0.$$

$$\therefore E[X_m - 0] = 0.$$

$\therefore (X_m - X_k)$ is uncorrelated with X_k . ($k < m$)

3) Y_1, Y_2, Y_3 are iid with $P(Y_i = 1) = p$.
and $P(Y_i = -1) = q = 1-p$, and $p \neq q \neq \frac{1}{2}$.

Fix $\mathcal{F}$, $\{a, b\}$ with $S_0 = 0$, and $S_n = Y_1 + Y_2 + \dots + Y_n$.

Define: $T = \min \{n : S_n = -a \text{ or } S_n = b\}$.

(a) Verify $M_n = S_n - n(p-q)$ & $M_n = \left(\frac{q}{p}\right)^{S_n}$ are martingales.

Solution

$$S_{n+1} = S_n + Y_{n+1}. \quad \text{AND} \quad E[Y_i] = (p-q) \quad (\forall i \in \mathbb{Z})$$

$$E[M_{n+1} | S_1, S_2, \dots, S_n] = E[S_{n+1} - (n+1)(p-q) | S_1, \dots, S_n].$$

Since S_n is known @ time $= n$.

$$E[S_{n+1} | S_1, \dots, S_n] = E[S_n + Y_{n+1} | S_1, S_2, \dots, S_n]$$

$$\Rightarrow E[S_n | S_1, \dots, S_n] + E[Y_{n+1}] = (S_n + (p-q))$$

$\therefore$ plugging it back, we get.

$$E[M_{n+1} | S_1, S_2, \dots, S_n] = S_n + (p-q) - (n+1)(p-q).$$

$$= S_n + (p-q) - n(p-q) - (p-q) = S_n - n(p-q) = M_n. \quad \text{Q.E.D.}$$

Again $S_{n+1} = S_n + Y_{n+1}$.

$$M_{n+1} = \left(\frac{q}{p}\right)^{S_{n+1}} = \left(\frac{q}{p}\right)^{S_n} \cdot \left(\frac{q}{p}\right)^{Y_{n+1}}.$$

$$E[M_{n+1} | \text{past upto } n] = \left(\frac{q}{p}\right)^{S_n} E\left[\left(\frac{q}{p}\right)^{Y_{n+1}} | \text{past}\right].$$

Now, Y_{n+1} is independent of past.

$$\therefore E[M_{n+1} | \text{past upto } n] = \left(\frac{q}{p}\right)^{S_n} \cdot E\left[\left(\frac{q}{p}\right)^{Y_{n+1}}\right].$$

$$E\left[\left(\frac{q}{p}\right)^{Y_{n+1}}\right] = p \cdot \left(\frac{q}{p}\right)^1 + q \cdot \left(\frac{q}{p}\right)^{-1}$$

$$= p \cdot \frac{q}{p} + q \cdot \left(\frac{p}{q}\right)$$

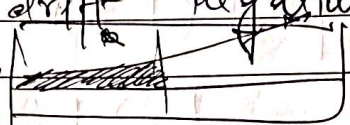
$$= p + q = 1$$

$$\therefore E[M_{n+1} | \text{past upto } n] = \left(\frac{q}{p}\right)^{S_n} = M_n$$

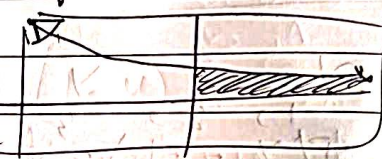
$$\therefore E[M_{n+1} | \text{past upto } n] = M_n \quad \leftarrow \text{martingale proved}.$$

- 4 (b) Let $r = q/p$.
- (*) Every +1 step, r shrinks value.
 - (*) Every -1 step, r boosts value.

(*) Therefore if $q > p$, $r > 1$ but since S_n will drift negatively, r^{S_n} over time be smaller.



(*) On the contrary, if $p > q$, $r < 1$, since S_n will drift positively, r^{S_n} over time again become smaller.



let-

$$u = P(S_T = b) \quad 1 - u = P(S_T = -a)$$

$\therefore$ At time T , walk must be @ one of the two boundary points whichever is hit earlier, $(S_T \in \{-a, b\})$.

$$E\left[\left(\frac{q}{p}\right)^{S_T}\right] = 1$$

$$\therefore u \left(\frac{q}{p}\right)^b + (1-u) \left(\frac{q}{p}\right)^{-a} = 1$$

$$u \left[\left(\frac{q}{p}\right)^b - \left(\frac{q}{p}\right)^{-a} \right] = 1 - \left(\frac{q}{p}\right)^{-a}$$

$$u = \frac{\left(\frac{q}{p}\right)^a - 1}{\left(\frac{q}{p}\right)^{a+b} - 1} = \frac{1 - \left(\frac{q}{p}\right)^{-a}}{\left(\frac{q}{p}\right)^{a+b} - 1}$$

(c) Find $E[\uparrow]$.

$$E[S_T - \tau(p-q)] = E[S_0] = 0. \quad (\text{martingale stop time})$$

$$E[S_T] - (p-q) E[\uparrow] = 0.$$

$$E[\uparrow] = \frac{E[S_T]}{p-q}$$

$$= \frac{b \cdot u + (1-a)(1-u)}{p-q}$$

$$= \frac{b \left[\frac{1 - (q/p)^a}{1 - (q/p)^{a+b}} \right] + (1-a) \left[1 - \frac{1 - (q/p)^a}{1 - (q/p)^{a+b}} \right]}{p-q}$$

$$E[\uparrow] = \frac{b \left[1 - (q/p)^a \right] - a (q/p)^a \left[1 - (q/p)^b \right]}{(p-q) \left[1 - (q/p)^{a+b} \right]}. \quad (\text{ans})$$